

Lesson 06 Demo 04

Creating Azure Traffic Manager

Objective: To create an Azure Traffic Manager to observe its working functionality by checking how high availability works with Azure Traffic Manager

Tools Required: Azure account with administrator access

Prerequisites: None

Steps to be followed:

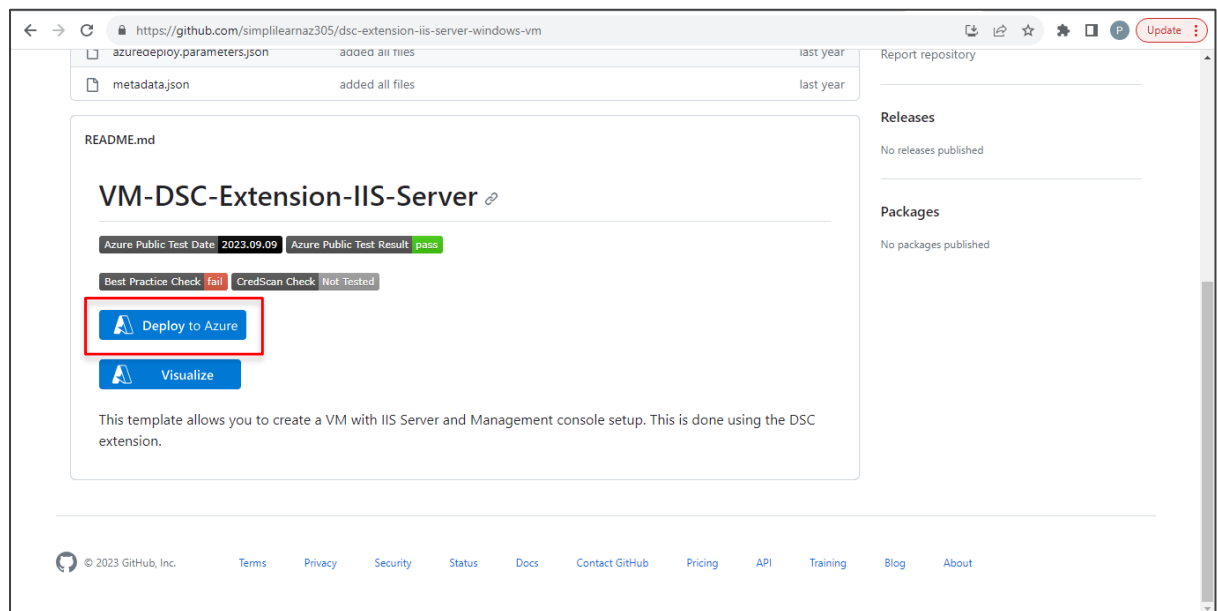
1. Create a virtual machine
2. Add DNS names to the virtual machine
3. Create a traffic manager
4. Test traffic manager

Step 1: Create a virtual machine

1.1 Log in to the Azure portal <https://portal.azure.com/>

1.2 Deploy the virtual machine to the Azure platform by clicking on **Deploy to Azure** after entering the given Simplilearn's GitHub link:

<https://github.com/simplilearnaz305/dsc-extension-iis-server-windows-vm>



1.3 In the Custom deployment page, provide the following details to create the first VM:
 create a resource group rg02, set Disk Type to StandardSSD_LRS, VM Name to iis first
 VM, Admin Username to Simpli@123, and Admin Password to Simplilearn@#1234.
 Click on **Review + create**

Home > Custom deployment ...

Deploy from a custom template

New! Deployment Stacks let you manage the lifecycle of your deployments. Try it now →

Basics Review + create

Template

Customized template 6 resources

Edit template Edit parameters Visualize

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * ① Simplilearn HOL 100648

Resource group * ① (New) rg02

Create new

Instance details

Region * ① East US

Disk Type ① StandardSSD_LRS

Previous Next Review + create

Home > Custom deployment ...

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Vm Name ① iis

Vm Size ① Standard_D2s_v3

Image SKU ① 2019-Datacenter

Admin Username * ① simpli@123

Admin Password * ①

Module File Path ① ContosoWebsite.ps1.zip

Configuration Function ① ContosoWebsite.ps1\ContosoWebsite

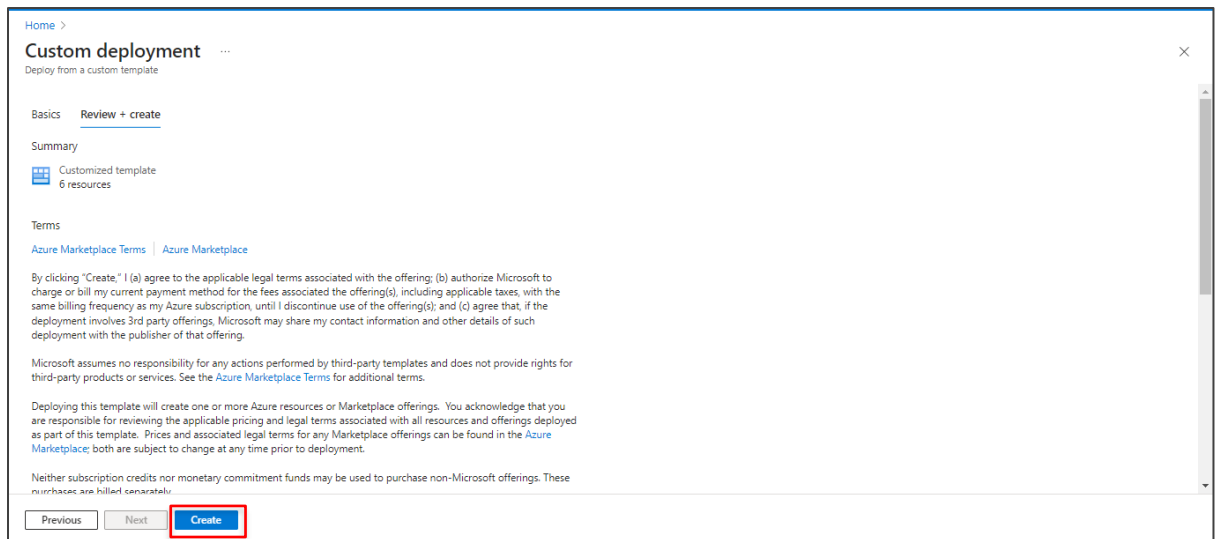
_artifacts Location ① [deployment].properties.templateLinkUri

_artifacts Location Sas Token ①

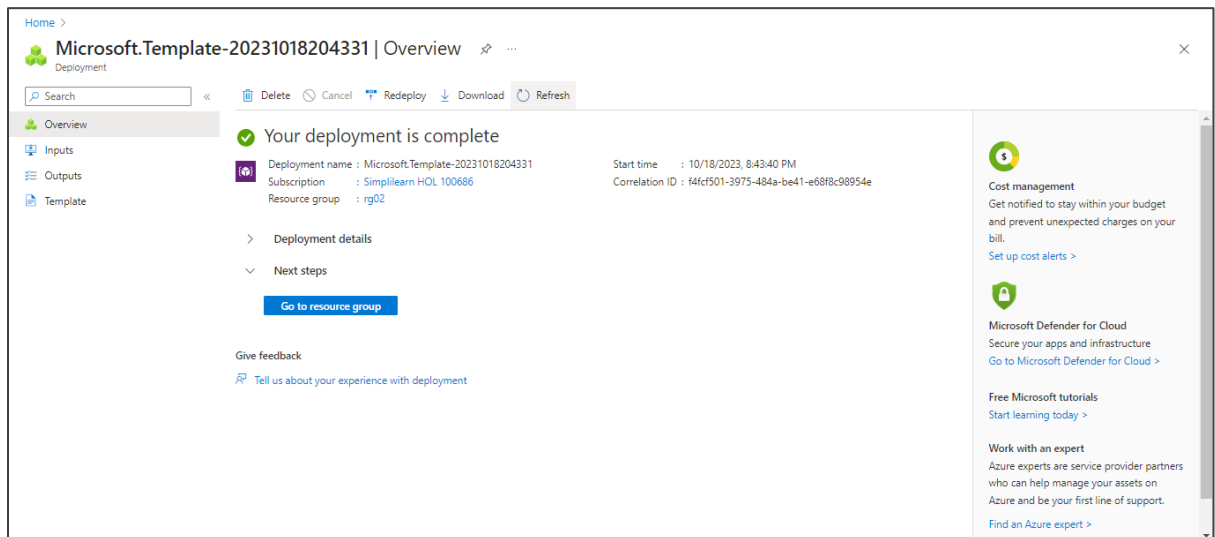
Location ① [resourceGroup().location]

Previous Next Review + create

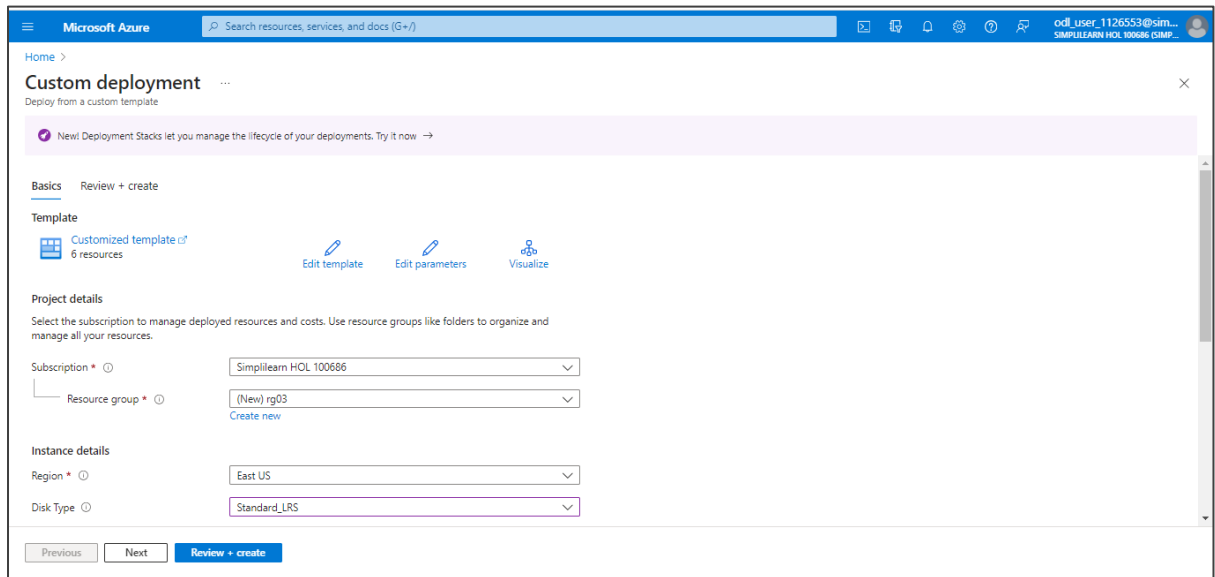
1.4 Click on Create



The output will appear as follows:



1.5 Provide the following details to create the second VM: create a resource group (rg03), set Disk Type to StandardSSD_LRS, VM Name to iis1 (second VM), Admin Username to **Simpli1234**, and Admin Password to **Simplilearn@#12345**. Click on **Review + create**



Microsoft Azure

Search resources, services, and docs (G+)

odl_user_1126553@sim...
SIMPILILEARN HOL 100686 CIMP...

Home >

Custom deployment

Deploy from a custom template

New! Deployment Stacks let you manage the lifecycle of your deployments. Try it now →

Basics Review + create

Template

Customized template of 6 resources

Edit template Edit parameters Visualize

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *

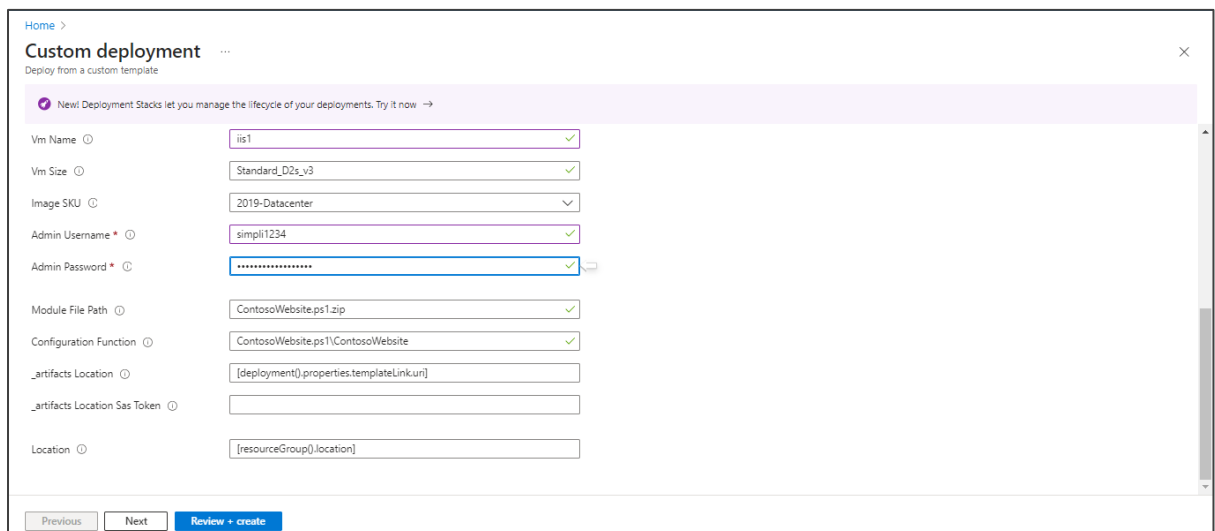
Resource group *
[Create new](#)

Instance details

Region *

Disk Type

[Previous](#) [Next](#) [Review + create](#)



Home >

Custom deployment

Deploy from a custom template

New! Deployment Stacks let you manage the lifecycle of your deployments. Try it now →

VM Name

VM Size

Image SKU

Admin Username *

Admin Password *

Module File Path

Configuration Function

_artifacts Location

_artifacts Location Sas Token

Location

[Previous](#) [Next](#) [Review + create](#)

1.6 Click on Create

Home > Microsoft.Template-20231017154746 | Overview >

Custom deployment

Deploy from a custom template

Basics **Review + create**

Summary

Custom template
6 resources

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Deploying this template will create one or more Azure resources or Marketplace offerings. You acknowledge that you are responsible for reviewing the applicable pricing and legal terms associated with all resources and offerings deployed as part of this template. Prices and associated legal terms for any Marketplace offerings can be found in the [Azure Marketplace](#); both are subject to change at any time prior to deployment.

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Previous Next **Create**

The output will appear as follows:

Microsoft Azure
Search resources, services, and docs (G+/)

Home >

Microsoft.Template-20231018210723 | Overview

Deployment

Search
Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

✓ Your deployment is complete

Deployment name : Microsoft.Template-20231018210723 Start time : 10/18/2023, 9:07:28 PM

Subscription : Simplilearn HOL 100686 Correlation ID : 2878df03-12ce-4bf0-831b-fddaa94e60ec

Resource group : rg03

> Deployment details

✓ Next steps

Go to resource group

Give feedback

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Deployment succeeded

Deployment 'Microsoft.Template-20231018210723' to resource group 'rg03' was successful.

[Pin to dashbo...](#) [Go to resource gr...](#)

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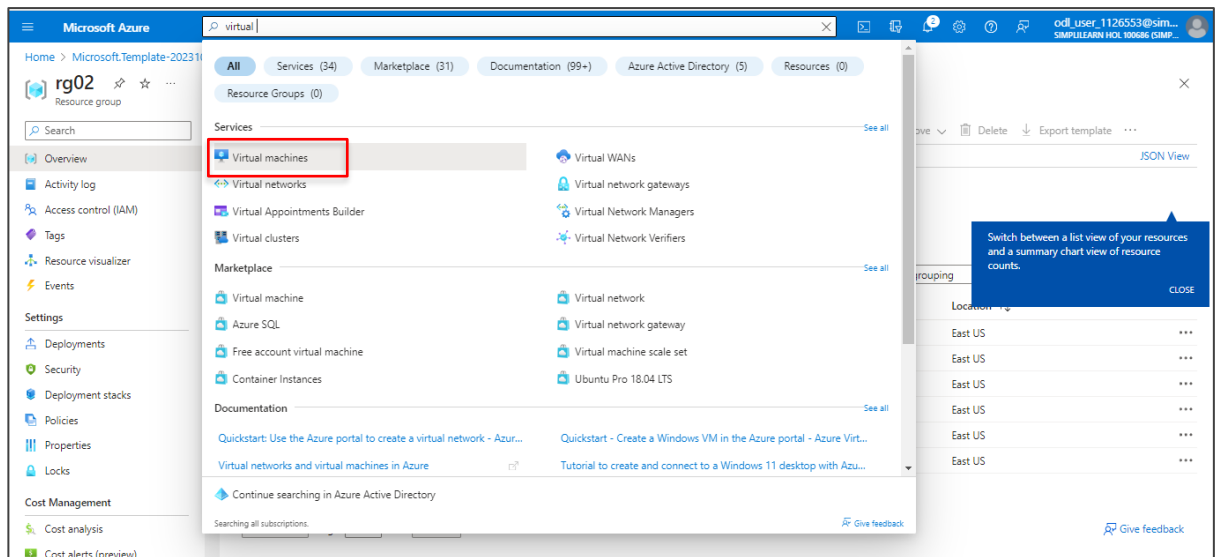
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Work with an expert

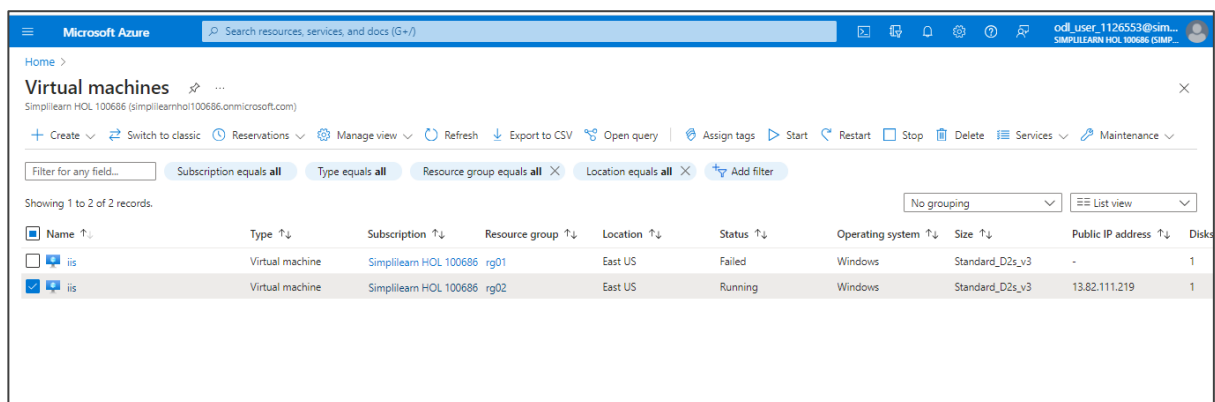
Azure experts are service provider partners who can help manage your assets on Azure and be your first line of support.

Step 2: Add DNS names to the virtual machine

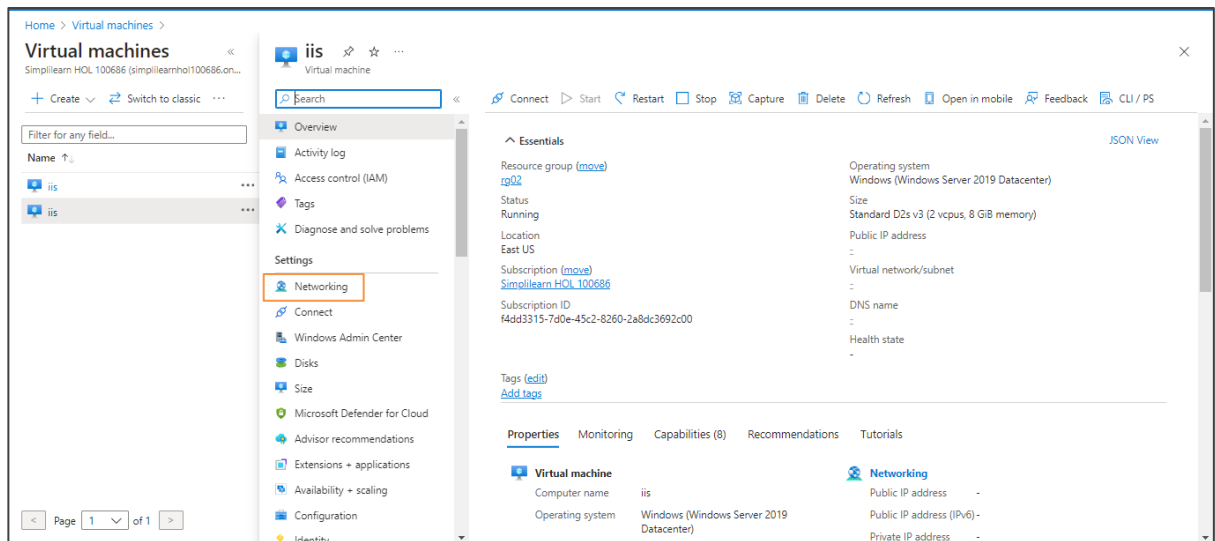
2.1 Search for Virtual machines in the search bar



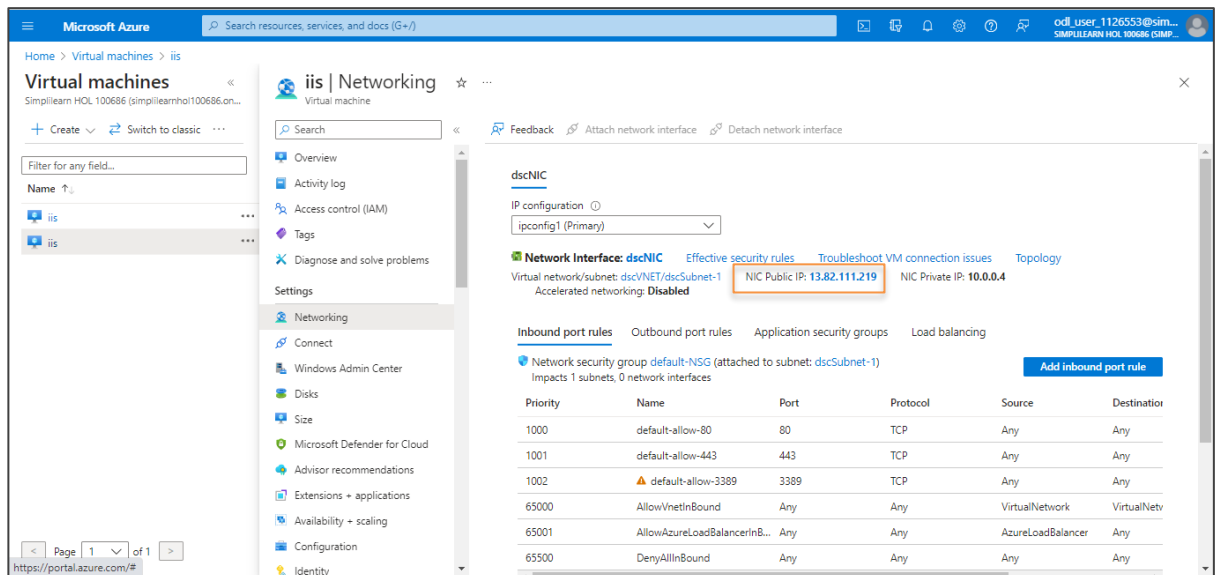
2.2 Open the first Virtual machine



2.3 Under **Settings**, select **Networking**



2.4 Select **NIC Public IP:13.82.111.219**



2.5 In the **DNS name label (optional)** field, enter a unique DNS name label (**myvirtualmachine**) and click on **Save**

The screenshot shows the 'dscPubIP' Configuration page in the Microsoft Azure portal. The left sidebar contains navigation links for Overview, Activity log, Access control (IAM), Tags, Settings, Configuration, Properties, Locks, Monitoring, Insights, Alerts, Metrics, Diagnostic settings, Logs, and Automation. The main content area shows the 'Configuration' tab with the following settings:

- IP address assignment: ☒ Dynamic ☐ Static
- IP address: 13.82.111.219
- Idle timeout (minutes): 4
- DNS name label (optional): myvirtualmachine

Below the settings, there is a message: "You can use the IP address as your 'A' DNS record or DNS label as your 'CNAME' record. [Learn more about adding a custom domain to this IP address](#)". There is also a section for 'Alias record sets' with a link to 'Create an alias record in Azure DNS' and a '+ Create alias record' button. At the bottom, there is a table with columns: Subscription, DNS zone, Name, Type, and TTL. The table currently shows 'No results.'.

The following page will appear as follows:

The screenshot shows the 'dscPubIP' Configuration page in the Microsoft Azure portal after saving changes. A success message is displayed at the top right: "Saved public IP address changes. Successfully saved configuration changes to public IP address 'dscPubIP'". The configuration settings are the same as in the previous screenshot, but the 'DNS name label (optional)' field now has a green checkmark next to it, indicating it has been saved.

2.6 Navigate to the second virtual machine

Name	Type	Subscription	Resource group	Location	Status	Operating system	Size	Public IP address	Disks
iis	Virtual machine	Simplilearn HOL 100686	rg01	East US	Failed	Windows	Standard_D2s_v3	-	1
iis	Virtual machine	Simplilearn HOL 100686	rg02	East US	Running	Windows	Standard_D2s_v3	13.82.111.219	1
iis1	Virtual machine	Simplilearn HOL 100686	RG03	East US	Running	Windows	Standard_D2s_v3	52.255.144.51	1

2.7 Under Settings, select Networking

The screenshot shows the 'Settings' page for the virtual machine 'iis1'. The 'Networking' option is highlighted in the left-hand menu. The main pane displays the following details:

- Essentials:**
 - Resource group: [RG03](#)
 - Status: Running
 - Location: East US
 - Subscription: [Simplilearn HOL 100686](#)
 - Subscription ID: f4dd3315-7d0e-45c2-8260-2a8dc3692c00
 - Tags: [\(edit\)](#) [Add tags](#)
- Properties:**
 - Virtual machine: **iis1**
 - Operating system: Windows (Windows Server 2019 Datacenter)
 - Public IP address: [52.255.144.51](#) (Network interface)
 - Virtual network/subnet: [dscVNET7/dscSubnet-1](#)
 - DNS name: [Not configured](#)
 - Health state: -

2.8 Select NIC Public IP:52.255.144.51

The screenshot shows the Microsoft Azure portal interface. On the left, the 'Virtual machines' section is expanded, and 'iis1' is selected. The 'Networking' tab is active for the 'iis1' virtual machine. Under the 'Network Interface' section, 'dscNIC' is selected. The 'Effective security rules' tab is open, showing a table of inbound port rules. The 'NIC Public IP' is highlighted as 52.255.144.51.

Priority	Name	Port	Protocol	Source	Destination
1000	default-allow-80	80	TCP	Any	Any
1001	default-allow-443	443	TCP	Any	Any
1002	default-allow-3389	3389	TCP	Any	Any
65000	AllowVnetInBound	Any	Any	VirtualNetwork	VirtualNetw
65001	AllowAzureLoadBalancerInB...	Any	Any	AzureLoadBalancer	Any
65500	DenyAllInBound	Any	Any	Any	Any

2.11 In the DNS name label (optional) field, enter a unique DNS name label (myvirtualmachine1) and click on Save

The screenshot shows the Microsoft Azure portal interface. On the left, the 'Public IP addresses' section is expanded, and 'dscPubIP' is selected. The 'Configuration' tab is active for the 'dscPubIP' public IP address. The 'DNS name label (optional)' field is highlighted and contains the value 'myvirtualmachine1'.

IP address assignment: ☒ Dynamic ☐ Static

IP address: 52.255.144.51

Idle timeout (minutes): 4

DNS name label (optional): myvirtualmachine1

You can use the IP address as your 'A' DNS record or DNS label as your 'CNAME' record. [Learn more about adding a custom domain to this IP address.](#)

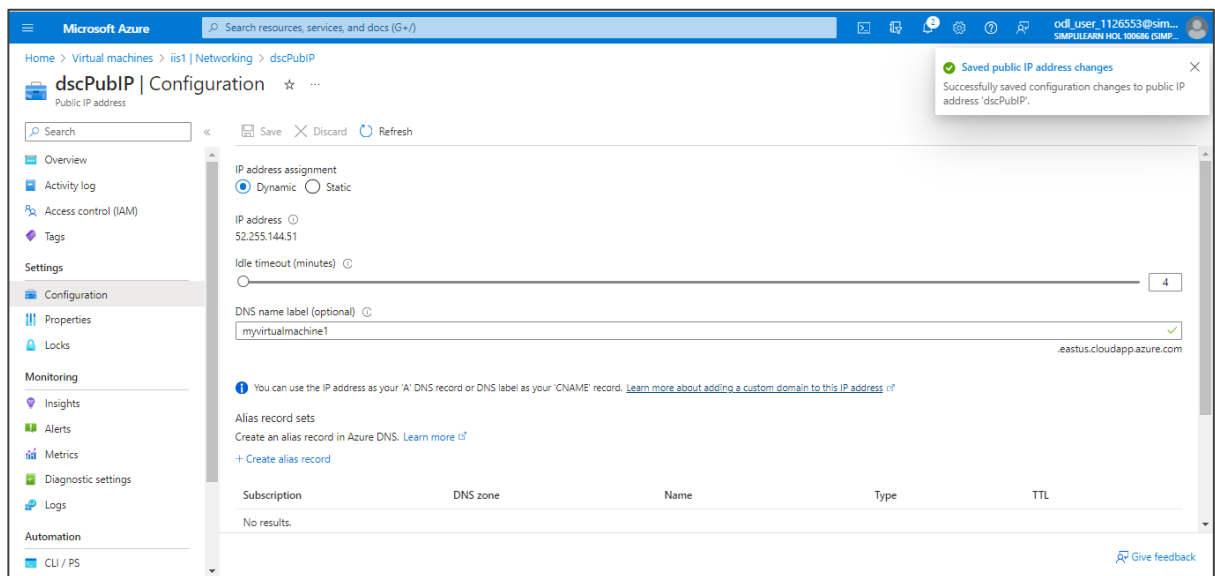
Alias record sets

Create an alias record in Azure DNS. [Learn more.](#)

+ Create alias record

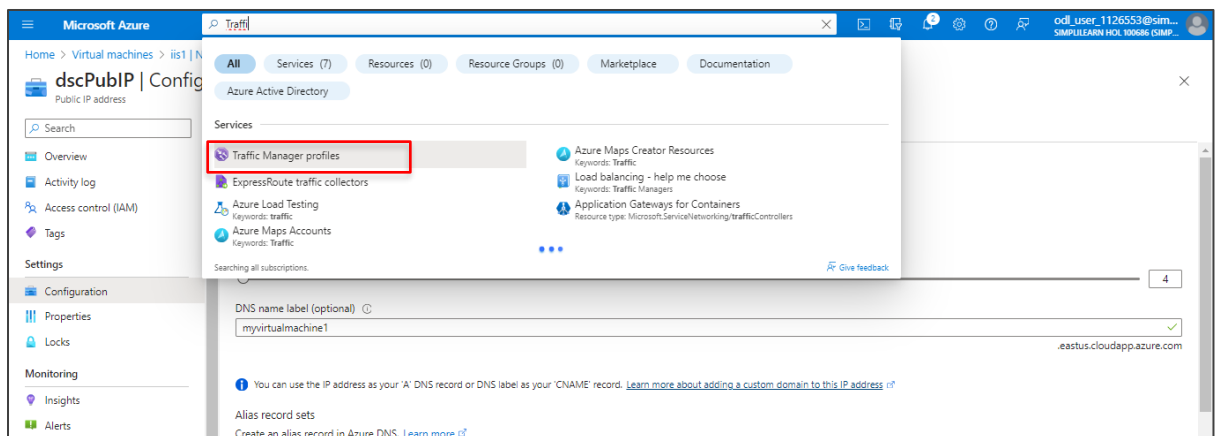
Subscription	DNS zone	Name	Type	TTL
No results.				

The following page will appear as follows:

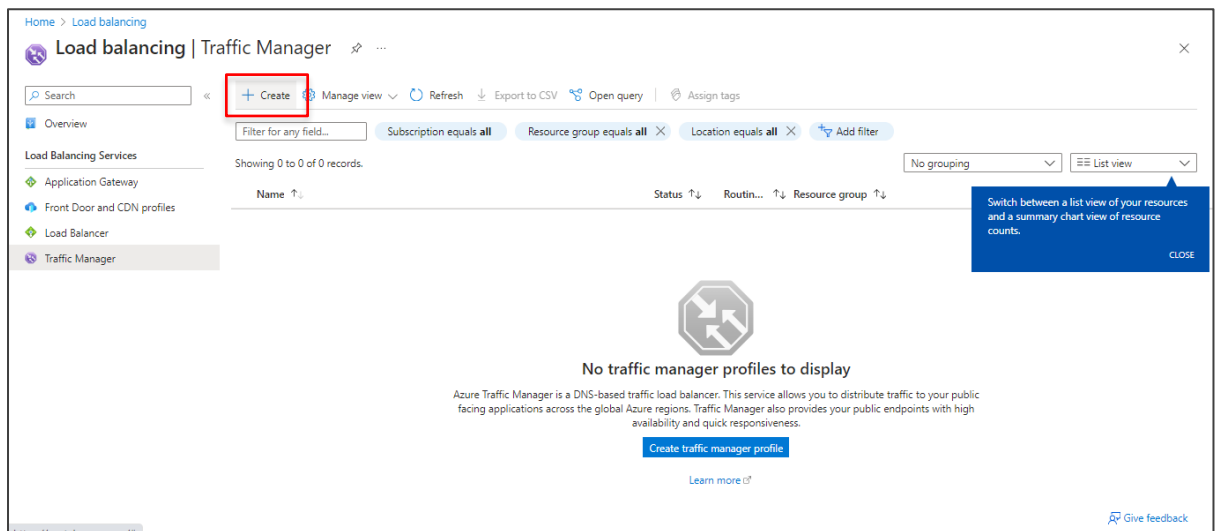


Step 3: Create a Traffic Manager

3.1 Search for and select Traffic Manager profiles



3.2 Click on + Create



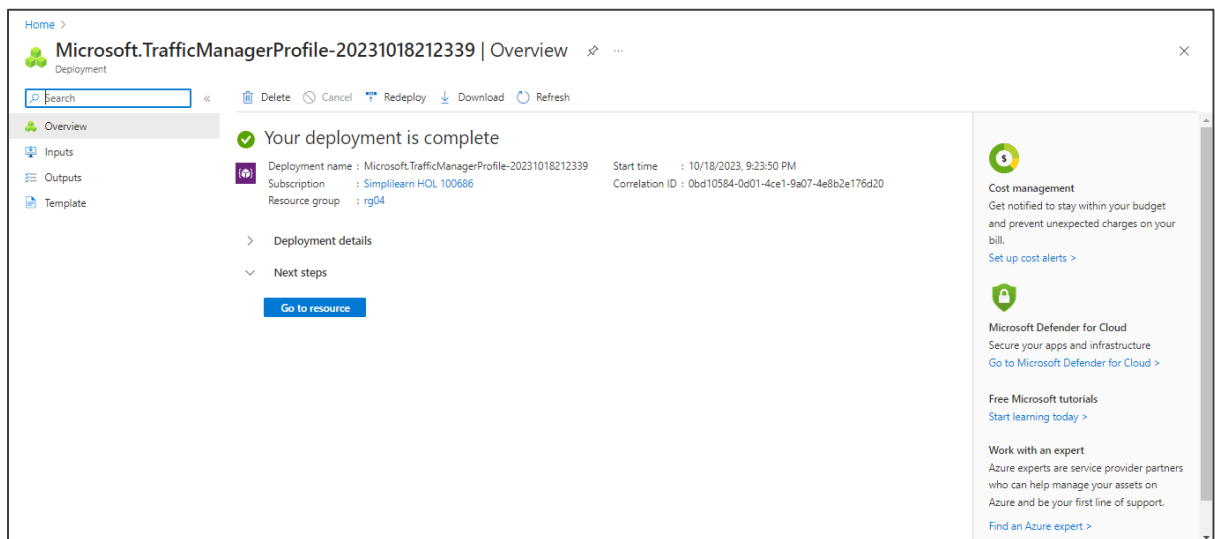
3.3 In the **Create Traffic Manager profile** page, enter the details as shown in the below screenshot. Now, click on **Create**.

The screenshot shows the 'Create Traffic Manager profile' form with the following details:

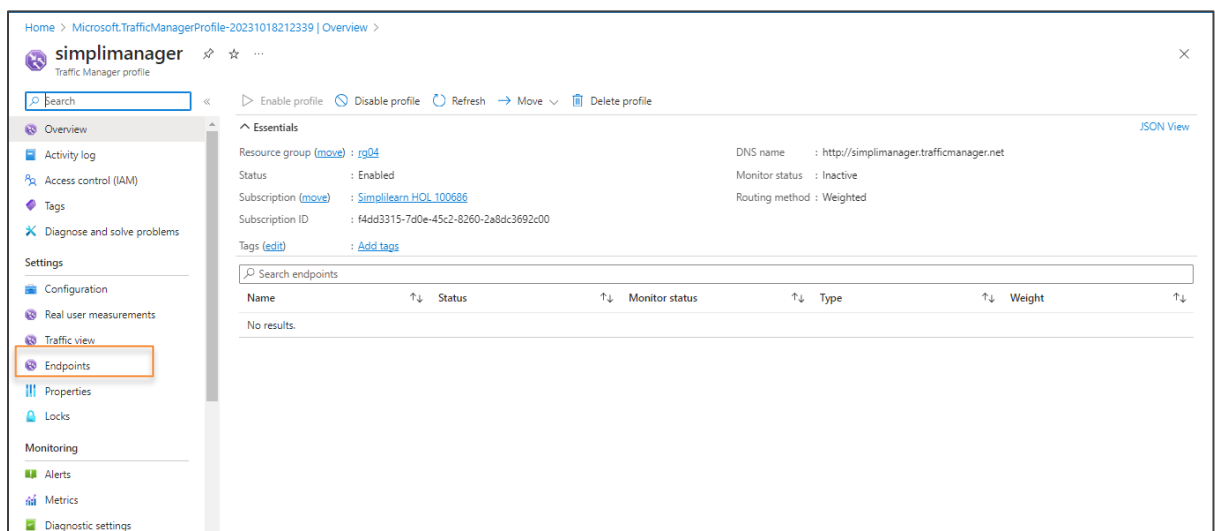
- Name:** simplimanager
- Routing method:** Weighted
- Subscription:** Simplilearn HOL 100686
- Resource group:** (New) rg04
- Resource group location:** East US

The form includes a 'Create' button and a link to 'Automation options'.

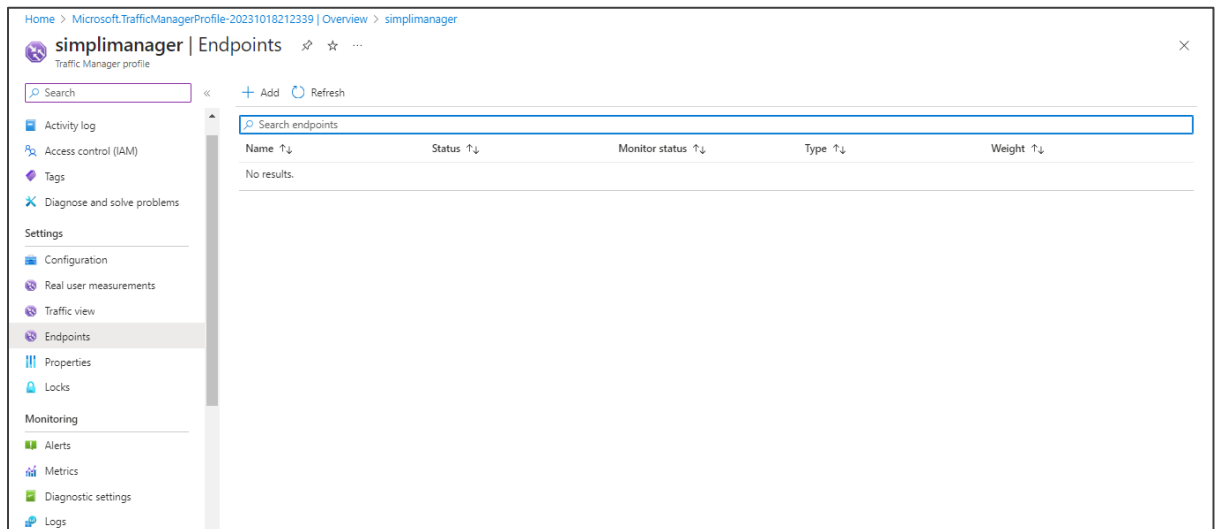
3.1 After the deployment is completed, click on **Go to resource**



3.2 Under **Settings**, select **Endpoints**



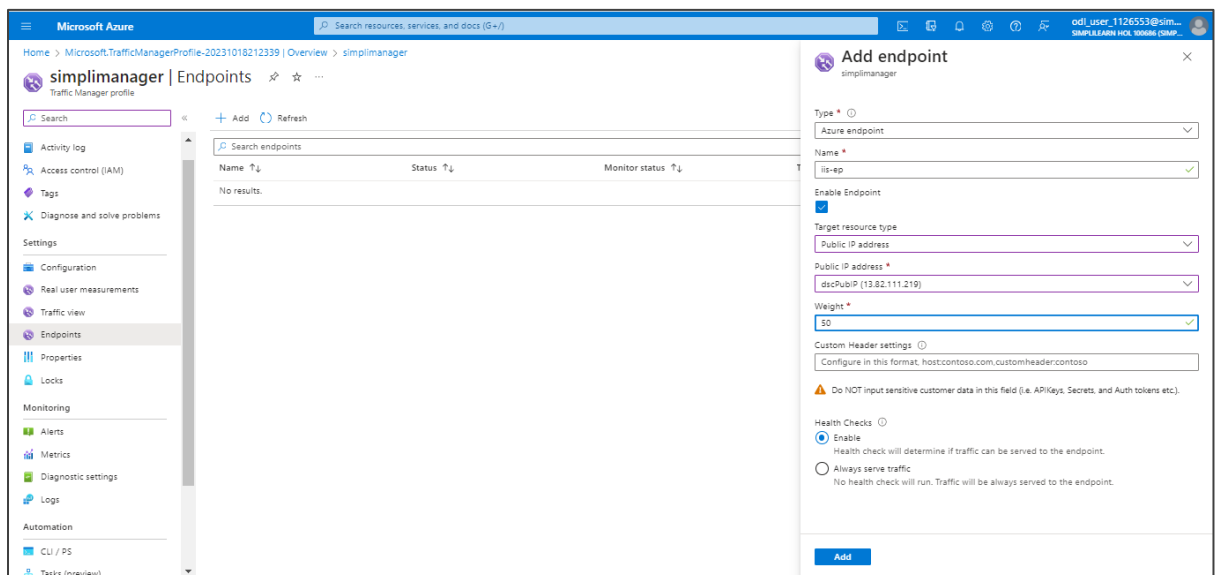
3.3 Click on + Add



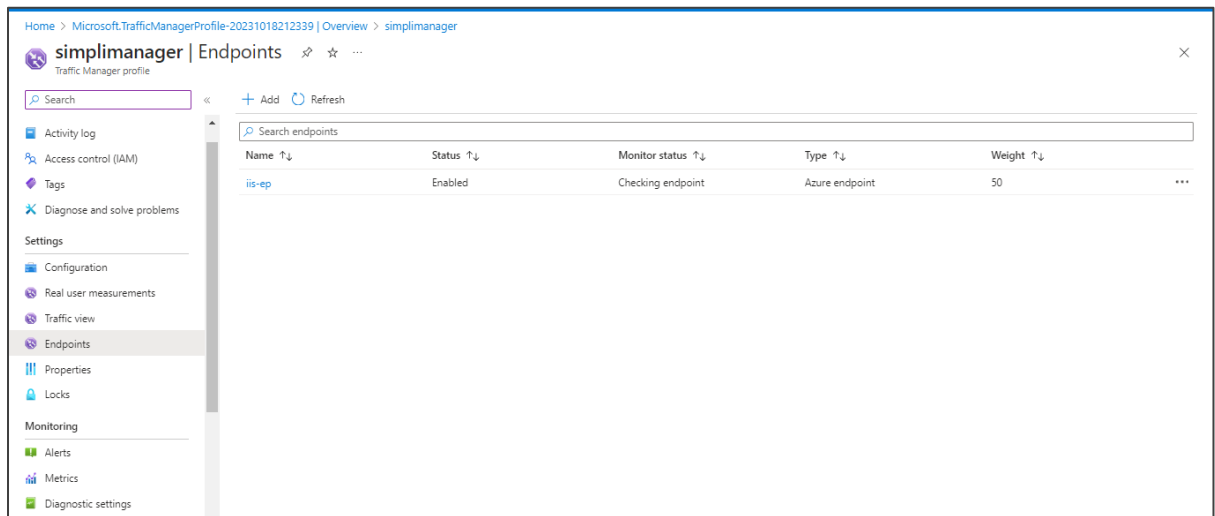
3.4 Provide the following values under Add endpoint:

- **Type:** Azure endpoint
- **Name:** iis-ep
- **Target resource type:** Public IP address
- **Public IP Address:** dscPubIP (13.82.111.219)
- **Weight:** 50

3.6 Click on Add



The endpoint for **iis-ep** (first VM) is shown below:



3.7 Provide the following values under **Add endpoint**:

Type: Azure endpoint

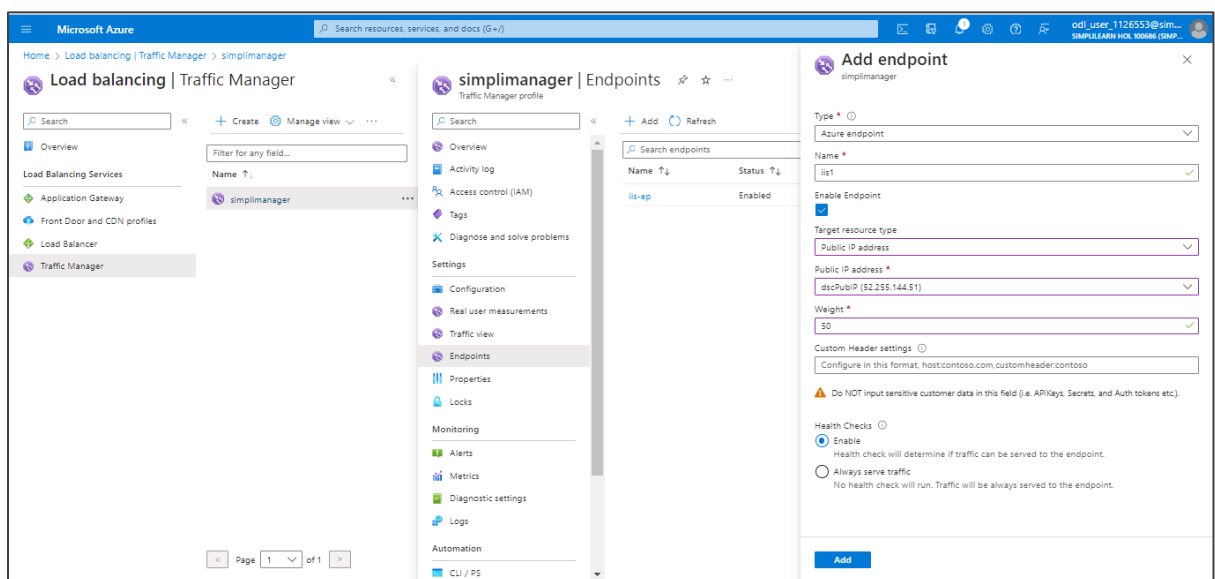
Name: iis1-ep

Target resource type: Public IP address

Public IP Address: dscPubIP (52.255.144.51)

Weight: 50

3.8 Click on **Add**



The endpoint for **iis1** (second VM) is shown below:

Name	Status	Monitor status	Type	Weight
iis-ep	Enabled	Online	Azure endpoint	50
iis1	Enabled	Checking endpoint	Azure endpoint	50

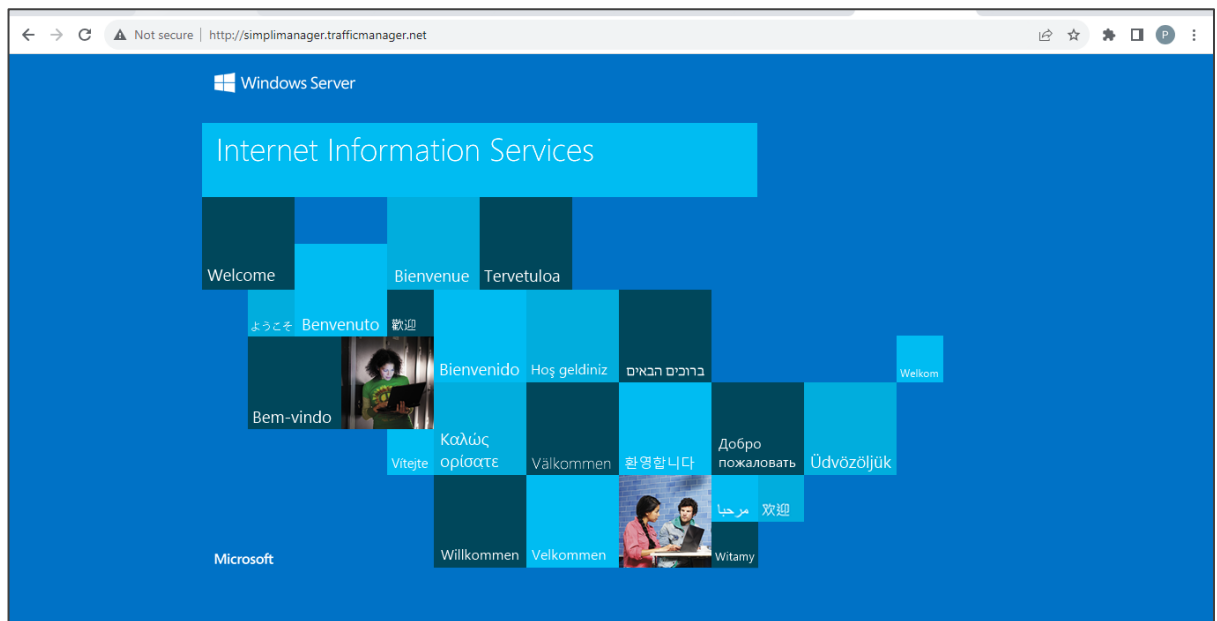
Step 4: Test traffic manager

4.1 Navigate to the **Traffic Manager**, and copy the **DNS name** from the Overview

DNS name: <http://simplimanager.trafficmanager.net> Copied

Name	Status	Monitor status	Type	Weight
iis-ep	Enabled	Online	Azure endpoint	50
iis1	Enabled	Online	Azure endpoint	50

4.2 Paste the URL in the browser



By following the above steps, you have successfully created an Azure Traffic Manager to observe its functionality and check how high availability works with it.